
Workshop on Multinomial- Processing-Tree (MPT) Modelling

The Software multiTree

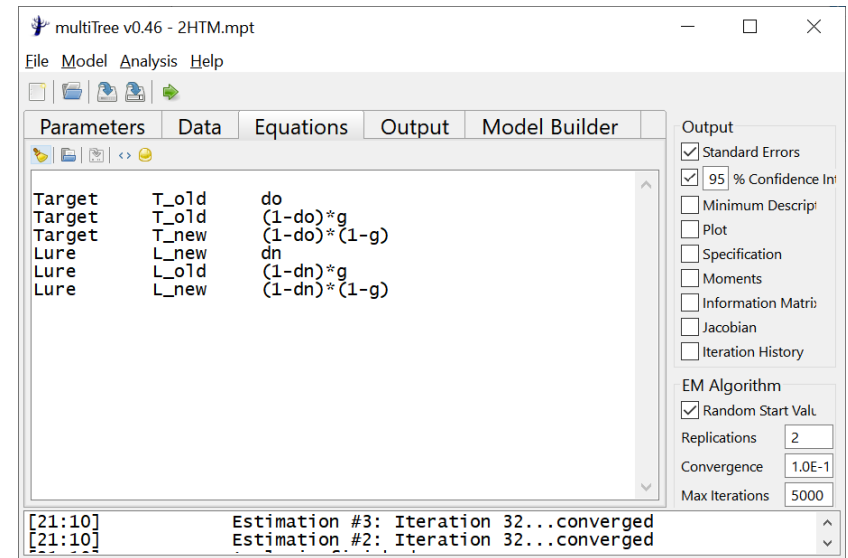


Prof. Dr. Daniel Heck

Software multiTree

multiTree (Moshagen, 2010)

- Specification, fitting and testing of MPT models
- Freeware
- Requires to install a Java Runtime Environment (JRE)
 - e.g., <https://www.java.com/en/download/>
- Download:
 - <https://www.sowi.uni-mannheim.de/erdfelder/forschung/software/multitree/>



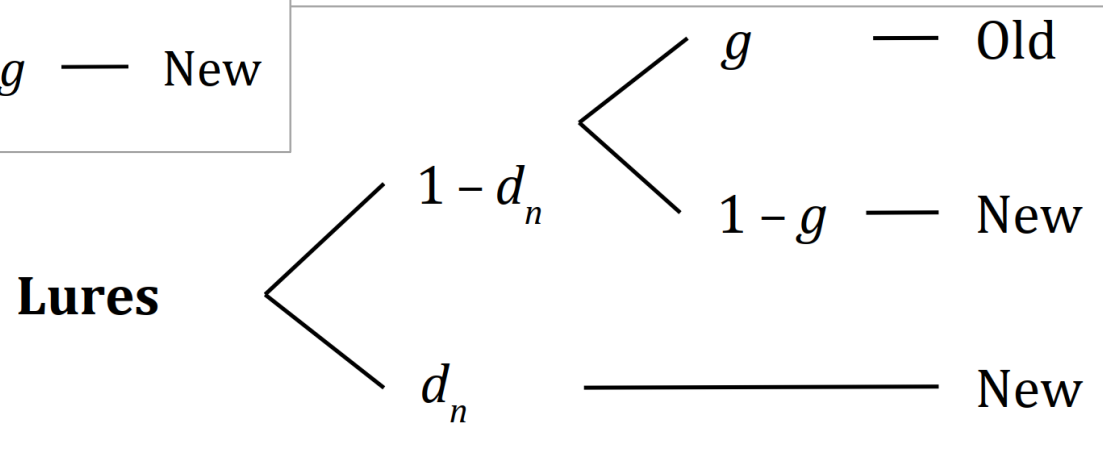
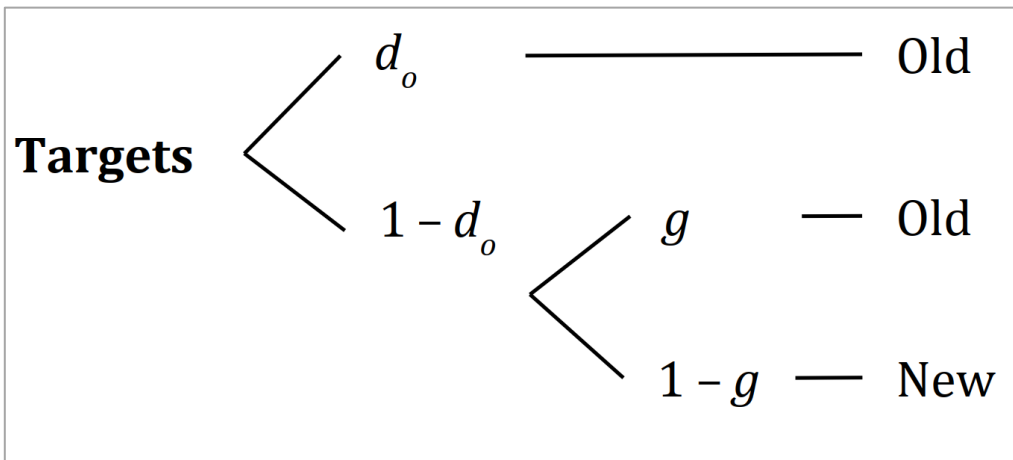
The Software multiTree

Overview

1. Define MPT model
2. Provide data
3. Check identifiability
4. Compare nested models

Application: Define MPT Model

1.) Open multiTree and provide the model equations of the **Two-High-Threshold Model**:

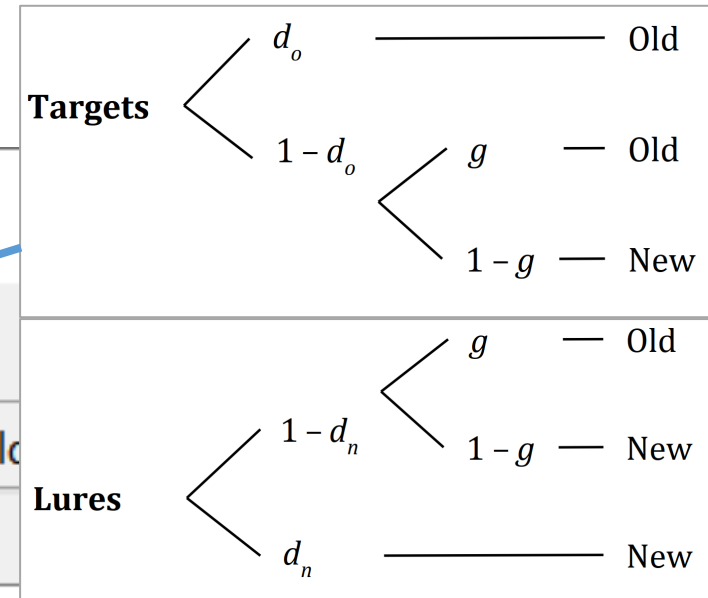


Two-High-Threshold Model (2HTM)

1.) Specification of an MPT model:

The screenshot shows the multiTree v0.46 software interface. The title bar reads "multiTree v0.46 - two-high-threshold-model.mpt". The menu bar includes "File", "Model", "Analysis", and "Help". The toolbar contains icons for file operations and a right-pointing arrow. Below the toolbar are tabs for "Parameters", "Data", "Equations", "Output", and "Model Build". The "Equations" tab is selected. Below the tabs is a toolbar with a bell icon, a folder icon, a document icon, a code icon, and a yellow circle icon. The main workspace displays a table with three columns: "Target", "Response category", and "Model equation".

Target	Response category	Model equation
Target	T_old	do
Target	T_old	$(1 - do) * g$
Target	T_new	$(1 - do) * (1 - g)$
Lure	L_new	dn
Lure	L_old	$(1 - dn) * g$
Lure	L_new	$(1 - dn) * (1 - g)$



graphical
model builder

condition

response category

model equation

The Software multiTree

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Application: Provide Data

2.) Provide the following **data / observed frequencies**:

	„old“	„new“
Target (old item)	65	35
Lure (new item)	13	87

Data

2.) Observed response frequencies:

	„old“	„new“
Old item	65	35
New item	13	87

The screenshot shows the multiTree v0.46 software interface. The title bar reads "multiTree v0.46 - two-high-threshold-model.mpt". The menu bar includes "File", "Model", "Analysis", and "Help". The toolbar contains icons for file operations and a green arrow pointing right. The "Data" tab is selected, showing a "Data Type" dropdown set to "Category Frequencies" and buttons for "Analyze" and "Batch Anal". The input fields are labeled as follows:

- Title: Example for MPT workshop
- T_old: 65
- T_new: 35
- L_old: 13
- L_new: 87

A blue arrow points from the "Data" tab to the table above. A red arrow points from the text "labels must be identical to the response categories in the model equations!" to the input fields for L_old and L_new.

labels must be identical to the response categories in the model equations!

The Software multiTree

Overview

1. Define MPT model
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Check Identifiability

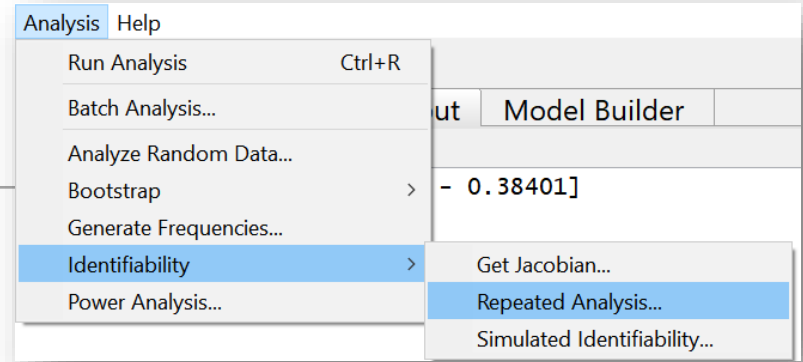
3.) Is the model **identifiable**?

The screenshot shows the multiTree v0.46 software interface. A blue arrow points from the word 'identifiable' in the question above to the 'Parameters' tab. The 'Parameters' tab is active, showing the 'Hierarchical Model Families' section with two unchecked checkboxes: 'Define current model as new baseline model (needs to be estimated before it can serve as a baseline)' and 'Compare current model against baseline model'. Below this, there are three rows of parameters: 'dn', 'do', and 'g', each with a 'free' dropdown and a '0.5' value field. To the right, the 'Specification' table is visible, with the 'Degrees of freedom' row highlighted by a red rectangle, showing a value of -1.

Specification	
Number of trees	2
Number of categories	4
Number of free categories	2
Number of parameters	3
Number of constrained parameters	0
Degrees of freedom	-1

On the right side, the 'Output' section has several checked options: 'Standard Errors', '95 % Confidence Intervals', 'Minimum Description Length', 'Plot', 'Specification', 'Moments', 'Information Matrix', 'Jacobian', and 'Iteration History'. The 'EM Algorithm' section has 'Random Start Values' checked.

Identifiability Checks in multiTree



- “Get Jacobian”

- Check whether the rank of the Jacobian matrix (for a random parameter θ) is equal to the number of free parameters

- “Repeated Analysis”

1. Estimate parameters multiple times for the same data
2. Check whether parameter estimates are stable/identical

- “Simulated Identifiability”

1. Draw random parameter vectors θ in Ω
2. Get expected category frequencies based on MPT model
3. Estimate the parameters of the MPT model
4. Check whether estimated & true parameters are identical

Application: Identifiability

- Example: Check identifiability via **repeated analysis**

multiTree v0.46 - 2HTM.mpt

File Model Analysis Help

Run Analysis Ctrl+R
Batch Analysis...
Analyze Random Data...
Bootstrap > - 0.38401]
Generate Frequencies...
Identifiability >
Power Analysis...

Get Jacobian...
Repeated Analysis...
Simulated Identifiability...

Repeated Analysis

Run	dn	do	g	Fit	AIC	BIC	Delta AIC	Delta BIC
1	0.20114	0.58197	0.16273	0.00000	212.76667	222.66162	2.00000	5.29832
2	0.57178	0.49743	0.30359	0.00000	212.76667	222.66162	2.00000	5.29832
3	0.50071	0.52679	0.26037	0.00000	212.76667	222.66162	2.00000	5.29832
4	0.62998	0.46043	0.35134	0.00000	212.76667	222.66162	2.00000	5.29832
5	0.70466	0.37482	0.44016	0.00000	212.76667	222.66162	2.00000	5.29832
6	0.71662	0.35336	0.45874	0.00000	212.76667	222.66162	2.00000	5.29832
7	0.62593	0.46358	0.34753	0.00000	212.76667	222.66162	2.00000	5.29832
8	0.67560	0.41594	0.40074	0.00000	212.76667	222.66162	2.00000	5.29832
9	0.78259	0.12949	0.59794	0.00000	212.76667	222.66162	2.00000	5.29832
10	0.62316	0.46567	0.34497	0.00000	212.76667	222.66162	2.00000	5.29832

Application: Identifiability

5.) Obtaining an identifiable model by equality constraints:

Parameters Data Equations Output Model Builder

Hierarchical Model Families

☐ Define current model as new baseline model (needs to be estimated before it can serve as a baseline).

☐ Compare current model against baseline model

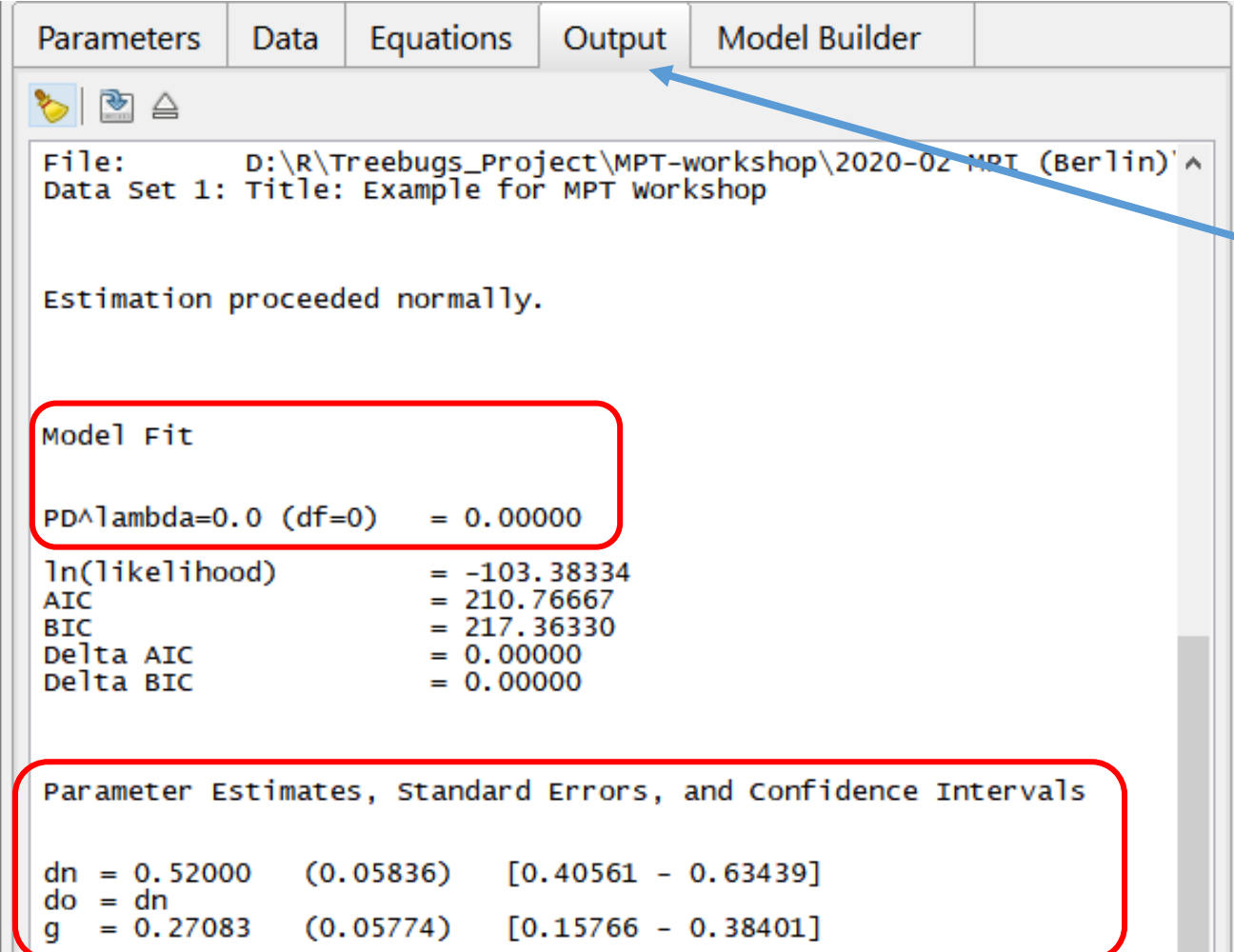
dn	free	0.52
do	= dn	0.52
g	free	0.27083

Specification

Number of trees	2
Number of categories	4
Number of free categories	2
Number of parameters	3
Number of constrained parameters	1
Degrees of freedom	0

Model Fitting: Output

6.) Results: Model fit and parameter estimates



Parameters Data Equations **Output** Model Builder

File: D:\R\Treebugs_Project\MPT-workshop\2020-02-MPT (Berlin)'^
Data Set 1: Title: Example for MPT workshop

Estimation proceeded normally.

Model Fit

PD λ lambda=0.0 (df=0) = 0.00000

ln(likelihood) = -103.38334
AIC = 210.76667
BIC = 217.36330
Delta AIC = 0.00000
Delta BIC = 0.00000

Parameter Estimates, Standard Errors, and Confidence Intervals

dn = 0.52000 (0.05836) [0.40561 - 0.63439]
do = dn
g = 0.27083 (0.05774) [0.15766 - 0.38401]

G² likelihood
ratio test

The Software multiTree

Overview

1. Define MPT model
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4. Comparing nested models

Comparing Nested Models

7.) Hypothesis testing: Does the constraint $g = .50$ hold?

→ Step 1: Define baseline model & fit model

When everything below is adjusted, it is necessary to fit the model once

Parameters Data Equations Output Model Builder

Hierarchical Model Families

☒ Define current model as new baseline model (needs to be estimated before it can serve as a baseline).

☐ Compare current model against baseline model

Specification		
Number of trees		2
Number of categories		4
Number of free categories		2
Number of parameters		3
Number of constrained parameters		1
Degrees of freedom		0

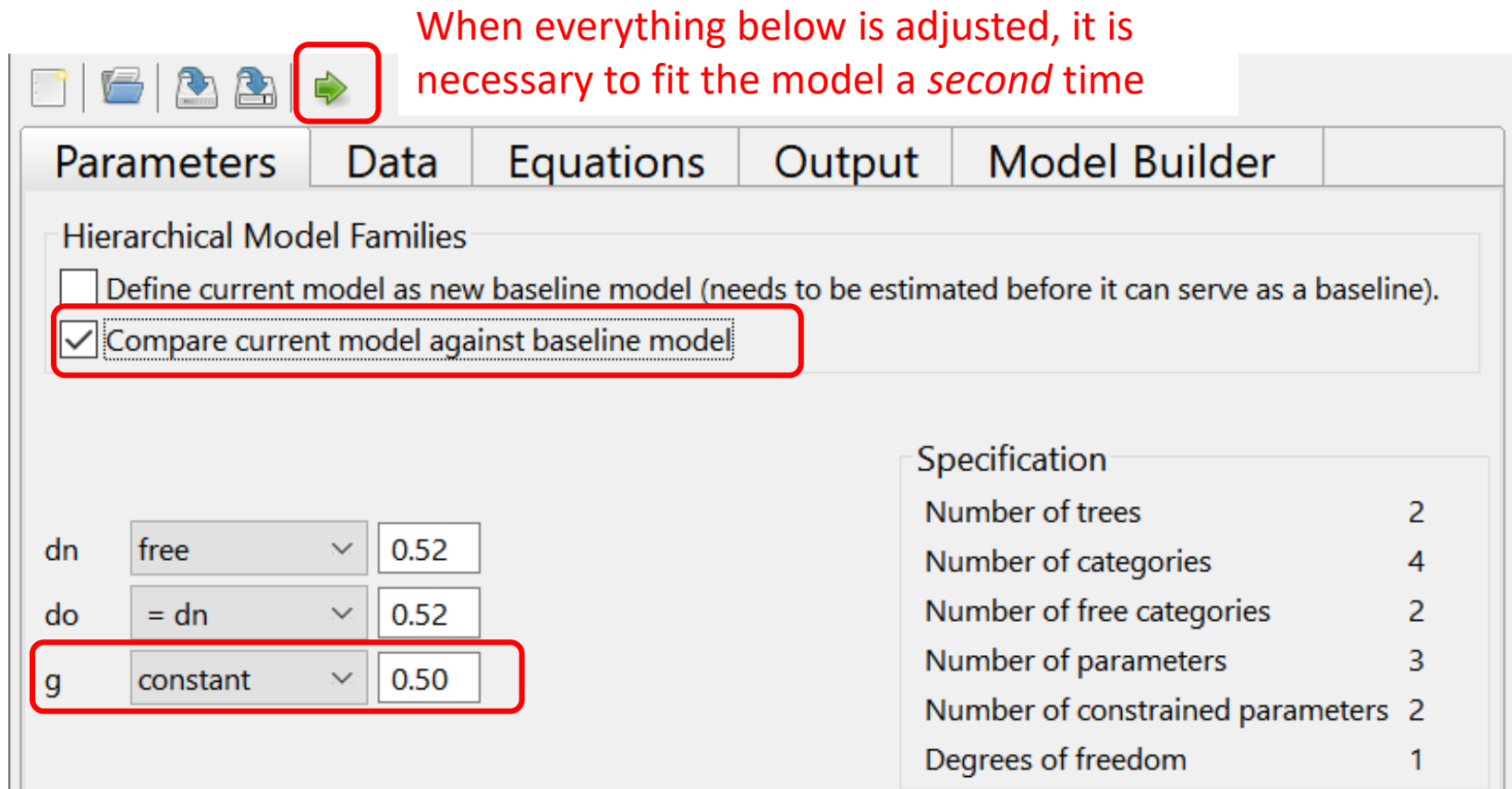
dn	free	0.52
do	= dn	0.52
g	free	0.2708

Comparing Nested Models

7.) Hypothesis testing: Does the constraint $g = .50$ hold?

→ Step 2: Compare current model to baseline

When everything below is adjusted, it is necessary to fit the model a *second* time



Parameters Data Equations Output Model Builder

Hierarchical Model Families

☐ Define current model as new baseline model (needs to be estimated before it can serve as a baseline).

☒ Compare current model against baseline model

dn free 0.52

do = dn 0.52

g constant 0.50

Specification

Number of trees	2
Number of categories	4
Number of free categories	2
Number of parameters	3
Number of constrained parameters	2
Degrees of freedom	1

Comparing Nested Models

- 7.) Hypothesis testing: Does the constraint $g = .50$ hold?

Likelihood
ratio test with
 ΔG^2 statistic

Parameters | Data | Equations | Output | Model Builder

Difference to Baseline Model (Difference = Current - Baseline)

PD^{lambda}=0.0 (df=1) = 13.66530 p = 0.00022

AIC difference = 11.66530

BIC difference = 8.36699

Ratio of AIC weights = 0.00292

Ratio of BIC weights = 0.01502

	Baseline	Current
dn	free	free
do	= dn	= dn
g	free	0.50

what are the parameters that are tested?

Parameter Estimates, Standard Errors, and Confidence Intervals

dn	= 0.52000	(0.06040)	[0.40162 - 0.63838]
do	= dn		
g	= 0.50000	(constant)	

what are the parameters
that are tested?

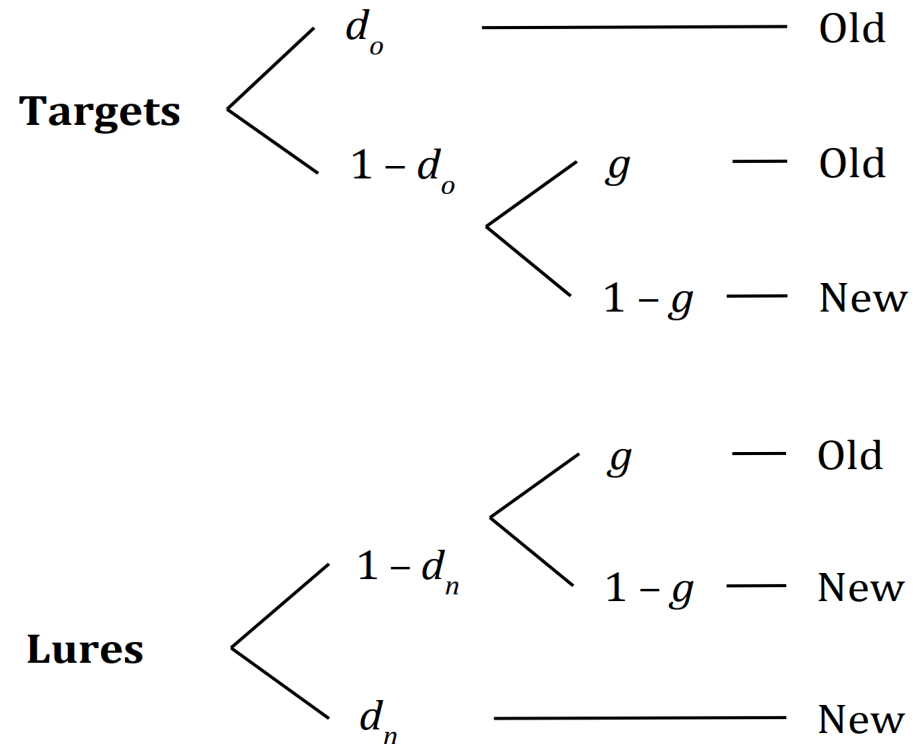
Practice with multiTree

Practice with multiTree

Preliminaries

1. Open [multiTree](#)
2. Open the file “[2HTM.mpt](#)”
3. The file contains:
 - Model equations
 - Data
 - Analysis results

Two-High Threshold Model (2HTM)



Practice with multiTree

1. Extend the 2HTM for **two base-rate conditions**:

		„old“	„new“
30% Targets	Target	65	35
	Lure	13	87
70% Targets	Target	83	17
	Lure	43	57

2. What are the **parameters** of the extended model?
3. Is the model **identifiable** with separate d_n and d_o ?
4. Fit the model to **both conditions jointly**.
5. **Test whether g differs** significantly between conditions.